Precalculus — Algebra I

Version 4 $- \epsilon$

by

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$\sigma o \phi \acute{\imath} \alpha$

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Preface

Thank you for your interest in our book, but more importantly, thank you for taking the time to read the Preface. I always read the Prefaces of the textbooks which I use in my classes because I believe it is in the Preface where I begin to understand the authors - who they are, what their motivation for writing the book was, and what they hope the reader will get out of reading the text. Pedagogical issues such as content organization and how professors and students should best use a book can usually be gleaned out of its Table of Contents, but the reasons behind the choices authors make should be shared in the Preface. Also, I feel that the Preface of a textbook should demonstrate the authors' love of their discipline and passion for teaching, so that I come away believing that they really want to help students and not just make money. Thus, I thank my fellow Preface-readers again for giving me the opportunity to share with you the need and vision which guided the creation of this book and passion which both Carl and I hold for Mathematics and the teaching of it.

Carl and I are natives of Northeast Ohio. We met in graduate school at Kent State University in 1997. I finished my Ph.D in Pure Mathematics in August 1998 and started teaching at Lorain County Community College in Elyria, Ohio just two days after graduation. Carl earned his Ph.D in Pure Mathematics in August 2000 and started teaching at Lakeland Community College in Kirtland, Ohio that same month. Our schools are fairly similar in size and mission and each serves a similar population of students. The students range in age from about 16 (Ohio has a Post-Secondary Enrollment Option program which allows high school students to take college

courses for free while still in high school.) to over 65. Many of the "non-traditional" students are returning to school in order to change careers. A majority of the students at both schools receive some sort of financial aid, be it scholarships from the schools' foundations, state-funded grants or federal financial aid like student loans, and many of them have lives busied by family and job demands. Some will be taking their Associate degrees and entering (or re-entering) the workforce while others will be continuing on to a four-year college or university. Despite their many differences, our students share one common attribute: they do not want to spend $200 on a College Algebra book.

The challenge of reducing the cost of textbooks is one that many states, including Ohio, are taking quite seriously. Indeed, state-level leaders have started to work with faculty from several of the colleges and universities in Ohio and with the major publishers as well. That process will take considerable time so Carl and I came up with a plan of our own. We decided that the best way to help our students right now was to write our own College Algebra book and give it away electronically for free. We were granted sabbaticals from our respective institutions for the Spring semester of 2009 and actually began writing the textbook on December 16, 2008. Using an open-source text editor called TexNicCenter and an open-source distribution of LaTeX called MikTex 2.7, Carl and I wrote and edited all of the text, exercises and answers and created all of the graphs (using Metapost within LaTeX) for Version $0.\overline{9}$ in about eight months. (We choose to create a text in only black and white to keep printing costs to a minimum for those students who prefer a printed edition. This somewhat Spartan page layout stands in sharp relief to the explosion of colors found in most other College Algebra texts, but neither Carl nor I believe the four-color print adds anything of value.) I used the book in three sections of College Algebra at Lorain County Community College in the Fall of 2009 and Carl's colleague, Dr. Bill Previts, taught a section of College Algebra at Lakeland with the book that semester as well. Students had the option of downloading the book as a .pdf file from our website www.stitz-zeager.com[1] or buying a low-cost printed version from our colleges' respective bookstores. (By

[1] http://www.stitz-zeager.com

giving this book away for free electronically, we end the cycle of new editions appearing every 18 months to curtail the used book market.) During Thanksgiving break in November 2009, many additional exercises written by Dr. Previts were added and the typographical errors found by our students and others were corrected. On December 10, 2009, Version $\sqrt{2}$ was released. The book remains free for download at our website and by using Lulu.com[2] as an on-demand printing service, our bookstores are now able to provide a printed edition for just under $19. Neither Carl nor I have, or will ever, receive any royalties from the printed editions. As a contribution back to the open-source community, all of the LaTeX files used to compile the book are available for free under a Creative Commons License on our website as well. That way, anyone who would like to rearrange or edit the content for their classes can do so as long as it remains free.

The only disadvantage to not working for a publisher is that we don't have a paid editorial staff. What we have instead, beyond ourselves, is friends, colleagues and unknown people in the open-source community who alert us to errors they find as they read the textbook. What we gain in not having to report to a publisher so dramatically outweighs the lack of the paid staff that we have turned down every offer to publish our book. (As of the writing of this Preface, we've had three offers.) By maintaining this book by ourselves, Carl and I retain all creative control and keep the book our own. We control the organization, depth and rigor of the content which means we can resist the pressure to diminish the rigor and homogenize the content so as to appeal to a mass market. A casual glance through the Table of Contents of most of the major publishers' College Algebra books reveals nearly isomorphic content in both order and depth. Our Table of Contents shows a different approach, one that might be labeled "Functions First." To truly use The Rule of Four, that is, in order to discuss each new concept algebraically, graphically, numerically and verbally, it seems completely obvious to us that one would need to introduce functions first. (Take a moment and compare our ordering to the classic "equations first, then the Cartesian Plane and THEN functions" approach seen in most of

[2] http://www.lulu.com/content/paperback-book/college-algebra/7513097

the major players.) We then introduce a class of functions and discuss the equations, inequalities (with a heavy emphasis on sign diagrams) and applications which involve functions in that class. The material is presented at a level that definitely prepares a student for Calculus while giving them relevant Mathematics which can be used in other classes as well. Graphing calculators are used sparingly and only as a tool to enhance the Mathematics, not to replace it. The answers to nearly all of the computational homework exercises are given in the text and we have gone to great lengths to write some very thought provoking discussion questions whose answers are not given. One will notice that our exercise sets are much shorter than the traditional sets of nearly 100 "drill and kill" questions which build skill devoid of understanding. Our experience has been that students can do about 15-20 homework exercises a night so we very carefully chose smaller sets of questions which cover all of the necessary skills and get the students thinking more deeply about the Mathematics involved.

Critics of the Open Educational Resource movement might quip that "open-source is where bad content goes to die," to which I say this: take a serious look at what we offer our students. Look through a few sections to see if what we've written is bad content in your opinion. I see this open-source book not as something which is "free and worth every penny", but rather, as a high quality alternative to the business as usual of the textbook industry and I hope that you agree. If you have any comments, questions or concerns please feel free to contact me at jeff@stitz-zeager.com or Carl at carl@stitz-zeager.com.

Jeff Zeager
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January 25, 2010

Chapter 1

Introduction to Functions

1.1 Functions and their Representations

1.2 Constant and Linear Functions

1.3 Absolute Value Functions

1.4 Quadratic Functions

Chapter 2

Polynomial Functions

2.1 Graphs of Polynomial Functions

2.2 The Remainder and Factor Theorems

2.3 Real Zeros of Polynomials

2.4 Complex Zeros and the Fundamental Theorem of Algebra

Chapter 3

Rational Functions

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3.2 Graphs of Rational Functions

3.3 Inequalities involving Rational Functions and Applications

Chapter 4

Root, Radical and Power Functions

4.1 Root and Radical Functions

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4.3 Equations and Inequalities involving Power Functions

Chapter 5

Further Topics on Functions

5.1 Graphs of Functions

5.2 Function Arithmetic

5.3 Function Composition

5.4 Transformations of Graphs

5.5 Relations and Implicit Functions

5.6 Inverse Functions

In Section **??**, we defined functions as processes. In this section, we seek to reverse, or 'undo' those processes. As in real life, we will find that some processes (like putting on socks and shoes) are reversible while some (like baking a cake) are not.

Consider the function $f(x) = 3x + 4$. Starting with a real number input x, we apply two steps in the following sequence: first we multiply the input by 3 and, second, we add 4 to the result.

To reverse this process, we seek a function g which will undo each of these steps and take the output from f, $3x + 4$, and return the input x. If we think of the two-step process of first putting on socks then putting on shoes, to reverse the process, we first take off the shoes and then we take off the socks. In much the same way, the function g should undo each step of f but in the opposite order. That is, the function g should first *subtract* 4 from the input x then *divide* the result by 3. This leads us to the formula $g(x) = \frac{x-4}{3}$.

Let's check to see if the function g does the job. If $x = 5$, then $f(5) = 3(5) + 4 = 15 + 4 = 19$. Taking the output 19 from f, we substitute it into g to get $g(19) = \frac{19-4}{3} = \frac{15}{3} = 5$, which is our original input to f. To check that g does the job for all x in the domain of f, we take the generic output from f, $f(x) = 3x + 4$, and substitute that into g. That is, we simplify $g(f(x)) = g(3x + 4) = \frac{(3x+4)-4}{3} = \frac{3x}{3} = x$, which is our original input to f. If we carefully examine the arithmetic as we simplify $g(f(x))$, we actually see g first 'undoing' the addition of 4, and then 'undoing' the multiplication by 3.

Not only does g undo f, but f also undoes g. That is, if we take the output from g, $g(x) = \frac{x-4}{3}$, and substitute that into f, we get $f(g(x)) = f\left(\frac{x-4}{3}\right) = 3\left(\frac{x-4}{3}\right) + 4 = (x - 4) + 4 = x$. Using the language of function composition developed in Section **??**, the statements $g(f(x)) = x$ and $f(g(x)) = x$ can be written as $(g \circ f)(x) = x$ and $(f \circ g)(x) = x$, respectively.[1] Abstractly, we

[1] At the level of functions, $g \circ f = f \circ g = I$, where I is the identity function as defined as $I(x) = x$ for all real numbers, x.

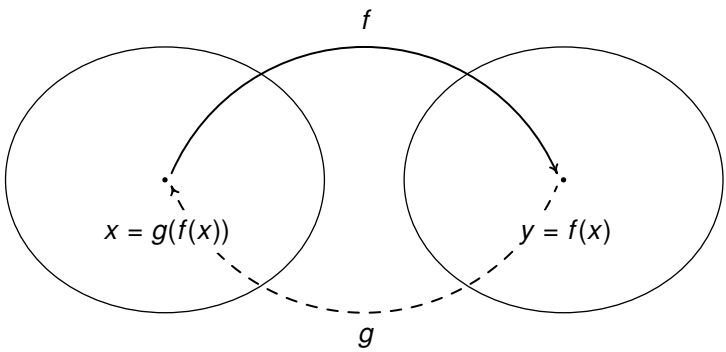


Figure 5.6.1

can visualize the relationship between f and g in Figure 5.6.1.

The main idea to get from the diagram is that g takes the outputs from f and returns them to their respective inputs, and conversely, f takes outputs from g and returns them to their respective inputs. We now have enough background to state the central definition of the section.

Definition 5.6.1. Suppose f and g are two functions such that

1. $(g \circ f)(x) = x$ for all x in the domain of f
   and
2. $(f \circ g)(x) = x$ for all x in the domain of g

then f and g are **inverses** of each other and the functions f and g are said to be **invertible**.

If we abstract one step further, we can express the sentiment in Definition 5.6.1 by saying that f and g are inverses if and only if $g \circ f = I_1$ and $f \circ g = I_2$ where I_1 is the identity function restricted[2] to the domain of f and I_2 is the identity function restricted to the domain of g.

In other words, $I_1(x) = x$ for all x in the domain of f and $I_2(x) = x$ for

[2]The identity function I, first introduced in Exercise **??** in Section **??** and mentioned in Theorem **??**, has a domain of all real numbers. Since the domains of f and g may not be all real numbers, we need the restrictions listed here.

all x in the domain of g. Using this description of inverses along with the properties of function composition listed in Theorem **??**, we can show that function inverses are unique.[3]

Suppose g and h are both inverses of a function f. By Theorem 5.6.1, the domain of g is equal to the domain of h, since both are the range of f. This means the identity function I_2 applies both to the domain of h and the domain of g. Thus $h = h \circ I_2 = h \circ (f \circ g) = (h \circ f) \circ g = I_1 \circ g = g$, as required.

We summarize the important properties of invertible functions in the following theorem.[4] Apart from introducing notation, each of the results below are immediate consequences of the idea that inverse functions map the outputs from a function f back to their corresponding inputs.

[3] In other words, invertible functions have exactly one inverse.

[4] In the interests of full disclosure, the authors would like to admit that much of the discussion in the previous paragraphs could have easily been avoided had we appealed to the description of a function as a set of ordered pairs. We make no apology for our discussion from a function composition standpoint, however, since it exposes the reader to more abstract ways of thinking of functions and inverses. We will revisit this concept again in Chapter **??**.

Theorem 5.6.1. Properties of Inverse Functions: Suppose f is an invertible function.

- There is exactly one inverse function for f, denoted f^{-1} (read 'f-inverse')
- The range of f is the domain of f^{-1} and the domain of f is the range of f^{-1}
- $f(a) = c$ if and only if $a = f^{-1}(c)$
  NOTE: In particular, for all y in the range of f, the solution to $f(x) = y$ is $x = f^{-1}(y)$.
- (a, c) is on the graph of f if and only if (c, a) is on the graph of f^{-1}
  NOTE: This means graph of $y = f^{-1}(x)$ is the reflection of the graph of $y = f(x)$ across $y = x$.[a]
- f^{-1} is an invertible function and $(f^{-1})^{-1} = f$.

[a] See Example **??** in Section **??** and Example **??** in Section **??**.

The notation f^{-1} is an unfortunate choice since you've been programmed since Elementary Algebra to think of this as $\frac{1}{f}$. This is most definitely *not* the case since, for instance, $f(x) = 3x + 4$ has as its inverse $f^{-1}(x) = \frac{x-4}{3}$, which is certainly different than $\frac{1}{f(x)} = \frac{1}{3x+4}$.

Why does this confusing notation persist? As we mentioned in Section **??**, the identity function I is to function composition what the real number 1 is to real number multiplication. The choice of notation f^{-1} alludes to the property that $f^{-1} \circ f = I_1$ and $f \circ f^{-1} = I_2$, in much the same way as $3^{-1} \cdot 3 = 1$ and $3 \cdot 3^{-1} = 1$.

Before we embark on an example, we demonstrate the pertinent parts of Theorem 5.6.1 to the inverse pair $f(x) = 3x + 4$ and $g(x) = f^{-1}(x) = \frac{x-4}{3}$. Suppose we wanted to solve $3x + 4 = 7$. Going through the usual machinations, we obtain $x = 1$.

If we view this equation as $f(x) = 7$, however, then we are looking for the input x corresponding to the output $f(x) = 7$. This is exactly the question f^{-1} was built to answer. In other words, the solution to $f(x) = 7$ is

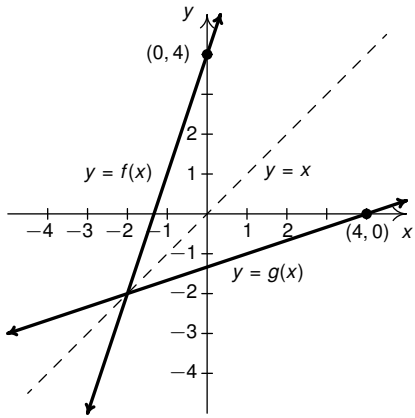


Figure 5.6.2: Graphs of inverse functions $y = f(x) = 3x + 4$ and $y = f^{-1}(x) = \dfrac{x-4}{3}$

$x = f^{-1}(7) = 1$. In other words, the formula $f^{-1}(x)$ encodes all of the algebra required to 'undo' what the formula $f(x)$ does to x. More generally, any time you have ever solved an equation, you have really been working through an inverse problem.

We also note the graphs of $f(x) = 3x+4$ and $g(x) = f^{-1}(x) = \frac{x-4}{3}$ are easily seen to be reflections across the line $y = x$ as seen below. In particular, note that the y-intercept $(0, 4)$ on the graph of $y = f(x)$ corresponds to the x-intercept on the graph of $y = f^{-1}(x)$. Indeed, the point $(0, 4)$ on the graph of $y = f(x)$ can be interpreted as $(0, 4) = (0, f(0)) = (f^{-1}(4), 4)$ just as the point $(4, 0)$ on the graph of $y = f^{-1}(x)$ can be interpreted as $(4, 0) = (4, f^{-1}(4)) = (f(0), 0)$. See Figure 5.6.2.

Example 5.6.1. For each pair of functions f and g below:

1. Verify each pair of functions f and g are inverses: (a) algebraically and (b) graphically.
2. Use the fact f and g are inverses to solve $f(x) = 5$ and $g(x) = -3$

- $f(x) = \sqrt[3]{x-1} + 2$ and $g(x) = (x-2)^3 + 1$

- $f(t) = \dfrac{2t}{t+1}$ and $g(t) = \dfrac{t}{2-t}$

Solution.

Solution for $f(x) = \sqrt[3]{x-1} + 2$ *and* $g(x) = (x-2)^3 + 1$.

1. (a) To verify $f(x) = \sqrt[3]{x-1} + 2$ and $g(x) = (x-2)^3 + 1$ are inverses, we appeal to Definition 5.6.1 and show $(g \circ f)(x) = x$ and $(f \circ g)(x) = x$ for all real numbers, x.

$$\begin{aligned}
(g \circ f)(x) &= g(f(x)) \\
&= g(\sqrt[3]{x-1} + 2) \\
&= [(\sqrt[3]{x-1} + 2) - 2]^3 + 1 \\
&= (\sqrt[3]{x-1})^3 + 1 \\
&= x - 1 + 1 \\
&= x \checkmark
\end{aligned}$$

$$\begin{aligned}
(f \circ g)(x) &= f(g(x)) \\
&= f((x-2)^3 + 1) \\
&= \sqrt[3]{[(x-2)^3 + 1] - 1} + 2 \\
&= \sqrt[3]{(x-2)^3} + 2 \\
&= x - 4 + 4 \\
&= x \checkmark
\end{aligned}$$

Since the root here, 3, is odd, Theorem **??** gives $(\sqrt[3]{x-1})^3 = x - 1$ and $\sqrt[3]{(x-2)^3} = x - 2$.

(b) To show f and g are inverses graphically, we graph $y = f(x)$ and $y = g(x)$ on the same set of axes and check to see if they are reflections about the line $y = x$.

The graph of $y = f(x) = \sqrt[3]{x-1} + 2$ appears in Figure 5.6.14 courtesy of Theorem **??** in Section **??**. The graph of $y = g(x) = (x-2)^3 + 1$ appears in Figure 5.6.4 thanks to Theorem **??** in Section **??**.

We can immediately see three pairs of corresponding points: $(0, 1)$ and $(1, 0)$, $(1, 2)$ and $(2, 1)$, $(2, 3)$ and $(3, 2)$. When graphed

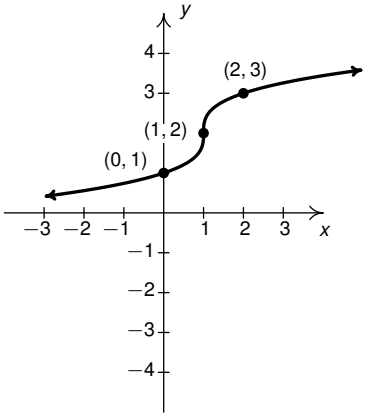


Figure 5.6.3: $y = f(x)$

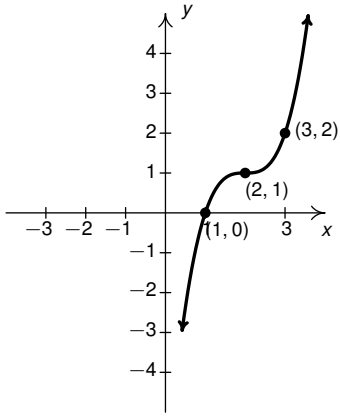


Figure 5.6.4: $y = g(x)$

on the same pair of axes, the two graphs certainly appear to be symmetric about the line $y = x$, as required. See Figure 5.6.5.

2. Since f and g are inverses, the solution to $f(x) = 5$ is $x = f^{-1}(5) = g(5) = (5-2)^3 + 1 = 28$. To check, we find $f(28) = \sqrt[3]{28-1} + 2 = \sqrt[3]{27} + 2 = 3 + 2 = 5$, as required.

   Likewise, the solution to $g(x) = -3$ is $x = g^{-1}(-3) = f(-3) = \sqrt[3]{(-3)-1} + 2 = 2 - \sqrt[3]{4}$. Once again, to check, we find $g(2 - \sqrt[3]{4}) = (2 - \sqrt[3]{4} - 2)^3 + 1 = (-\sqrt[3]{4})^3 + 1 = -4 + 1 = -3$.

Solution for $f(t) = \dfrac{2t}{t+1}$ *and* $g(t) = \dfrac{t}{2-t}$.

1. (a) Note the domain of f excludes $t = -1$ and the domain of g excludes $t = 2$. Hence, when simplifying $(g \circ f)(t)$ and $(f \circ g)(t)$, we tacitly assume $t \neq -1$ and $t \neq 2$, respectively.

$$\begin{aligned}
(g \circ f)(t) &= g(f(t)) \\
&= g\left(\frac{2t}{t+1}\right) \\
&= \frac{\dfrac{2t}{t+1}}{2 - \dfrac{2t}{t+1}} \\
&= \frac{\dfrac{2t}{}}{}
\end{aligned}$$

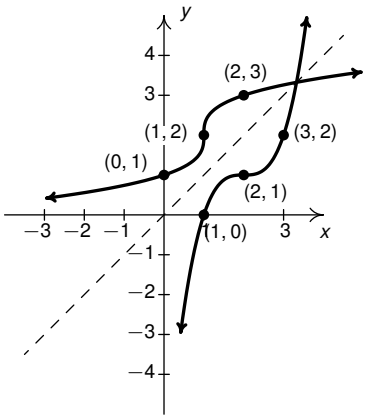


Figure 5.6.5: $y = f(x)$, $y = g(x)$, $y = x$

$$
\begin{aligned}
(f \circ g)(t) &= f(g(t)) \\
&= f\left(\frac{t}{2-t}\right) \\
&= \frac{2\left(\frac{t}{2-t}\right)}{\left(\frac{t}{2-t}\right)+1} \\
&= \frac{2\left(\frac{t}{2-t}\right)}{\left(\frac{t}{2-t}\right)+1} \cdot \frac{(2-t)}{(2-t)} \\
&= \frac{2t}{t+(1)(2-t)} \\
&= \frac{2t}{t+2-t)} \\
&= \frac{2t}{2} \\
&= t \checkmark
\end{aligned}
$$

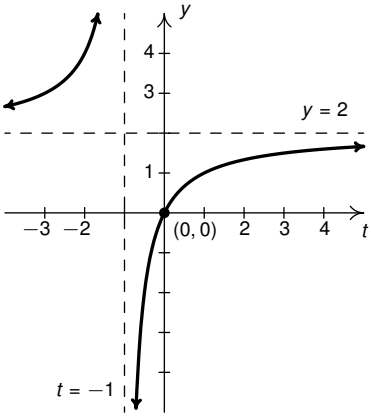


Figure 5.6.6: $y = f(t)$

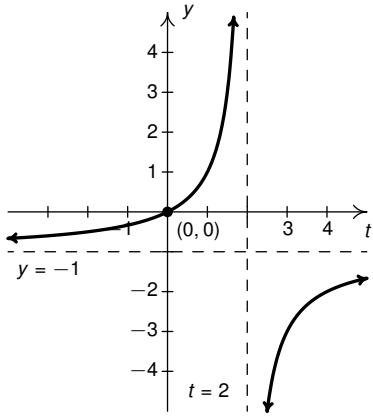


Figure 5.6.7: $y = g(t)$

(b) We graph $y = f(t)$ and $y = g(t)$ using the techniques discussed in Sections **??** and **??**. See Figure 5.6.6 and Figure 5.6.7.

We find the graph of f has a vertical asymptote $t = -1$ and a horizontal asymptote $y = 2$. Corresponding to the *vertical* asymptote $t = -1$ on the graph of f, we find the graph of g has a *horizontal* asymptote $y = -1$.

Likewise, the *horizontal* asymptote $y = 2$ on the graph of f corresponds to the *vertical* asymptote $t = 2$ on the graph of g. Both graphs share the intercept $(0, 0)$. When graphed together on the same set of axes, the graphs of f and g do appear to be symmetric about the line $y = t$. See Figure 5.6.8.

2. Don't let the fact that f and g in this case were defined using the independent variable, 't' instead of 'x' deter you in your efforts to solve $f(x) = 5$. Remember that, ultimately, the function f here is the *process* represented by the formula $f(t)$, and is the same process (with the same inverse!) regardless of the letter used as the independent variable. Hence, the solution to $f(x) = 5$ is $x = f^{-1}(1) = g(5)$. We get $g(5) = \frac{5}{2-5} = -\frac{5}{3}$.

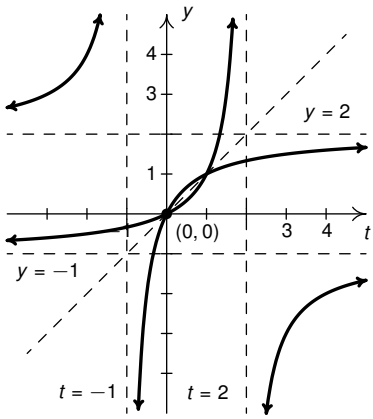


Figure 5.6.8: $y = f(t)$, $y = g(t)$, and $y = t$

To check, we find $f\left(-\frac{5}{3}\right) = \left(-\frac{10}{3}\right) / \left(-\frac{2}{3}\right) = 5$. Similarly, we solve $g(x) = -3$ by finding $x = g^{-1}(-3) = f(-3) = \frac{-6}{-2} = 3$. Sure enough, we find $g(3) = \frac{3}{2-3} = -3$. □

We now investigate under what circumstances a function is invertible. As a way to motivate the discussion, we consider $f(x) = x^2$. A likely candidate for the inverse is the function $g(x) = \sqrt{x}$. However, $(g \circ f)(x) = g(f(x)) = \sqrt{x^2} = |x|$, which is not equal to x unless $x \geq 0$.

For example, when $x = -2$, $f(-2) = (-2)^2 = 4$, but $g(4) = \sqrt{4} = 2$. That is, g failed to return the input -2 from its output 4. Instead, g matches the output 4 to a *different* input, namely 2, which satisfies $f(2) = 4$. This is shown schematically in Figure 5.6.9.

We see from the diagram that since both $f(-2)$ and $f(2)$ are 4, it is impossible to construct a *function* which takes 4 back to *both* $x = 2$ and $x = -2$ since, by definition, a function can match 4 with only *one* number.

In general, in order for a function to be invertible, each output can come from only *one* input. Since, by definition, a function matches up each input to only *one* output, invertible functions have the property that they match one input to one output and vice-versa. We formalize this concept below.

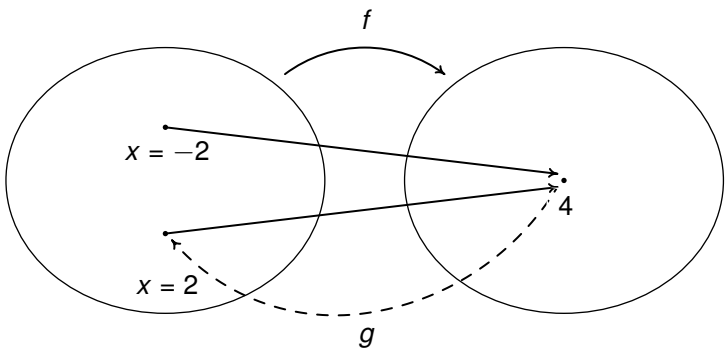


Figure 5.6.9

Definition 5.6.2. A function f is said to be **one-to-one** if whenever $f(a) = f(b)$, then $a = b$.

Note that an equivalent way to state Definition 5.6.2 is that a function is one-to-one if *different* inputs go to *different* outputs. That is, if $a \neq b$, then $f(a) \neq f(b)$.

Before we solidify the connection between invertible functions and one-to-one functions, we take a moment to see what goes wrong graphically when trying to find the inverse of $f(x) = x^2$.

Per Theorem 5.6.1, the graph of $y = f^{-1}(x)$, if it exists, is obtained from the graph of $y = x^2$ by reflecting $y = x^2$ about the line $y = x$. Procedurally, this is accomplished by interchanging the x and y coordinates of each point on the graph of $y = x^2$. Algebraically, we are swapping the variables 'x' and 'y' which results in the equation $x = y^2$ whose graph is shown in Figure 5.6.10.

We see immediately the graph of $x = y^2$ fails the Vertical Line Test, Theorem **??**. In particular, the vertical line $x = 4$ intersects the graph at two points, $(4, -2)$ and $(4, 2)$ meaning the relation described by $x = y^2$ matches the x-value 4 with two different y-values, -2 and 2.

Note that the *vertical* line $x = 4$ and the points $(4, \pm 2)$ on the graph of

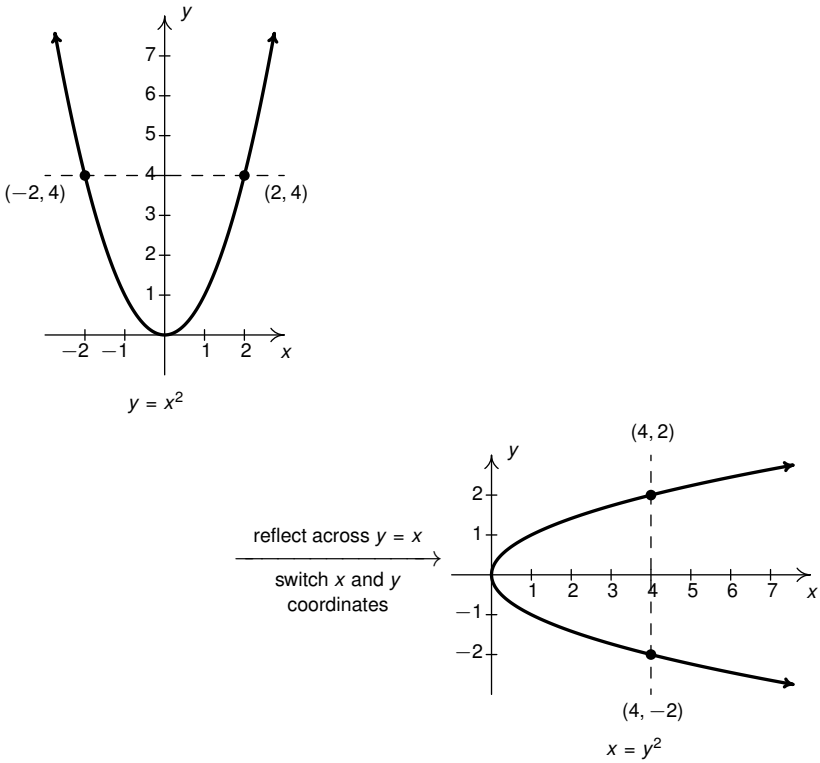

y
7
6
5
4
3
2
1
(−2, 4)
(2, 4)
−2 −1 1 2 x
y = x²
(4, 2)
y
2
1
reflect across y = x
switch x and y
coordinates
1 2 3 4 5 6 7 x
−1
−2
(4, −2)
x = y²

Figure 5.6.10

$x = y^2$ correspond to the *horizontal* line $y = 4$ and the points $(\pm 2, 4)$ on the graph of $y = x^2$ which brings us right back to the concept of one-to-one. The fact that both $(-2, 4)$ and $(2, 4)$ are on the graph of f means $f(-2) = f(2) = 4$. Hence, f takes different inputs, -2 and 2, to the same output, 4, so f is not one-to-one.

Recall the Horizontal Line Test from Exercise **??** in Section **??**. Applying that result to the graph of f we say the graph of f 'fails' the Horizontal Line Test since the horizontal line $y = 4$ intersects the graph of $y = x^2$ more than once. This means that the equation $y = x^2$ does not represent x is not a function of y.

Said differently, the Horizontal Line Test detects when there is at least one y-value (4) which is matched to more than one x-value (± 2). In other words, the Horizontal Line Test can be used to detect whether or not a function is one-to-one.

So, to review, $f(x) = x^2$ is not invertible, not one-to-one, and its graph fails the Horizontal Line Test. It turns out that these three attributes: being invertible, one-to-one, and having a graph that passes the Horizontal Line Test are mathematically equivalent. That is to say if one if these things is true about a function, then they all are; it also means that, as in this case, if one of these things *isn't* true about a function, then *none* of them are. We summarize this result in the following theorem.

Theorem 5.6.2. Equivalent Conditions for Invertibility:
For a function f, either all of the following statements are true or none of them are:

- f is invertible.
- f is one-to-one.
- The graph of f passes the Horizontal Line Test.[a]

[a] i.e., no horizontal line intersects the graph more than once.

To prove Theorem 5.6.2, we first suppose f is invertible. Then there is a function g so that $g(f(x)) = x$ for all x in the domain of f. If $f(a) = f(b)$, then $g(f(a)) = g(f(b))$. Since $g(f(x)) = x$, the equation $g(f(a)) = g(f(b))$

reduces to $a = b$. We've shown that if $f(a) = f(b)$, then $a = b$, proving f is one-to-one.

Next, assume f is one-to-one. Suppose a horizontal line $y = c$ intersects the graph of $y = f(x)$ at the points (a, c) and (b, c). This means $f(a) = c$ and $f(b) = c$ so $f(a) = f(b)$. Since f is one-to-one, this means $a = b$ so the points (a, c) and (b, c) are actually one in the same. This establishes that each horizontal line can intersect the graph of f at most once, so the graph of f passes the Horizontal Line Test.

Last, but not least, suppose the graph of f passes the Horizontal Line Test. Let c be a real number in the range of f. Then the horizontal line $y = c$ intersects the graph of $y = f(x)$ just *once*, say at the point $(a, c) = (a, f(a))$. Define the mapping g so that $g(c) = g(f(a)) = a$. The mapping g is a *function* since each horizontal line $y = c$ where c is in the range of f intersects the graph of f only *once*. By construction, we have the domain of g is the range of f and that for all x in the domain of f,$g(f(x)) = x$. We leave it to the reader to show that for all x in the domain of g, $f(g(x)) = x$, too.

Hence, we've shown: first, if f invertible, then f is one-to-one; second, if f is one-to-one, then the graph of f passes the Horizontal Line Test; and third, if f passes the Horizontal Line Test, then f is invertible. Hence if f is satisfies any one of these three conditions, we can show f must satisfy the other two.[5]

We put this result to work in the next example.

Example 5.6.2. Determine if the following functions are one-to-one: (a) analytically using Definition 5.6.2 and (b) graphically using the Horizontal Line Test. For the functions that are one-to-one, graph the inverse.

1. $f(x) = x^2 - 2x + 4$
2. $g(t) = \dfrac{2t}{1-t}$
3. $F = \{(-1, 1), (0, 2), (1, -3), (2, 1)\}$
4. $G = \{(t^3 + 1, 2t) \mid t \text{ is a real number.}\}$

[5]For example, if we know f is one-to-one, we showed the graph of f passes the HLT which, in turn, guarantees f is invertible.

Solution.

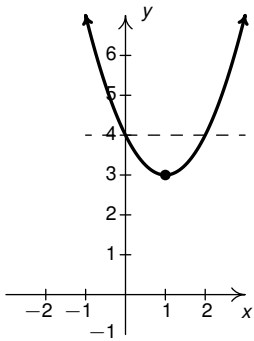
1. (a) To determine whether or not f is one-to-one analytically, we assume $f(a) = f(b)$ and work to see if we can deduce $a = b$.

$$\begin{aligned} f(a) &= f(b) \\ a^2 - 2a + 4 &= b^2 - 2b + 4 \\ a^2 - 2a &= b^2 - 2b \\ a^2 - b^2 - 2a + 2b &= 0 \\ (a+b)(a-b) - 2(a-b) &= 0 \\ (a-b)((a+b) - 2) &= 0 \\ a - b = 0 \quad &\text{or} \quad a + b - 2 = 0 \\ a = b \quad &\text{or} \quad a = 2 - b \end{aligned}$$

As we work our way through the problem, we encounter a quadratic equation. We rewrite the equation so it equals 0 and factor by grouping. We get $a = b$ as one possibility, but we also get the possibility that $a = 2 - b$. This suggests that f may not be one-to-one. Taking $b = 0$, we get $a = 0$ or $a = 2$. Since $f(0) = 4$ and $f(2) = 4$, we have two different inputs with the same output, proving f is neither one-to-one nor invertible.

(b) We note that f is a quadratic function and we graph $y = f(x)$ using the techniques presented in Section **??** in Figure 5.6.11. We see the graph fails the Horizontal Line Test quite often - in particular, crossing the line $y = 4$ at the points $(0, 4)$ and $(2, 4)$.

2. (a) We begin with the assumption that $g(a) = g(b)$ for a, b in the domain of g (That is, we assume $a \neq 1$ and $b \neq 1$.) Through our work below, we deduce $a = b$, proving g is one-to-one.



$y = f(x)$

Figure 5.6.11

$$\begin{aligned} g(a) &= g(b) \\ \frac{2a}{1-a} &= \frac{2b}{1-b} \\ 2a(1-b) &= 2b(1-a) \\ 2a - 2ab &= 2b - 2ba \\ 2a &= 2b \\ a &= b \checkmark \end{aligned}$$

(b) We graph $y = g(t)$ in Figure 5.6.12 using the procedure outlined in Section **??**. We find the sole intercept is $(0, 0)$ with asymptotes $t = 1$ and $y = -2$. Based on our graph, the graph of g appears to pass the Horizontal Line Test, verifying g is one-to-one.

Since g is one-to-one, g is invertible. Even though we do not have a *formula* for $g^{-1}(t)$, we can nevertheless sketch the graph of $y = g^{-1}(t)$ by reflecting the graph of $y = g(t)$ across $y = t$. See Figure 5.6.13.

Corresponding to the *vertical* asymptote $t = 1$ on the graph of g, the graph of $y = g^{-1}(t)$ will have a *horizontal* asymptote $y = 1$. Similarly, the *horizontal* asymptote $y = -2$ on the graph

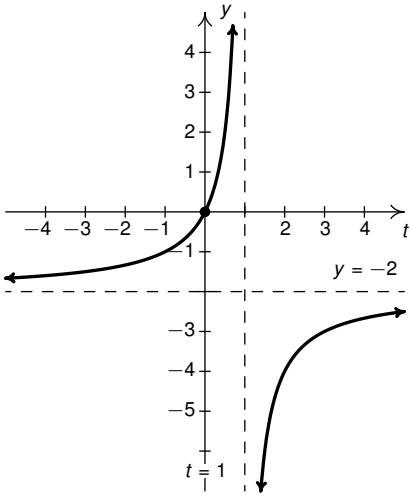


Figure 5.6.12: $y = g(t)$

of g corresponds to a *vertical* asymptote $t = -2$ on the graph of g^{-1}. The point $(0, 0)$ remains unchanged when we switch the t and y coordinates, so it is on both the graph of g and g^{-1}.

3. (a) The function F is given to us as a set of ordered pairs. Recall each ordered pair is of the form $(a, F(a))$. Since $(-1, 1)$ and $(2, 1)$ are both elements of F, this means $F(-1) = 1$ and $F(2) = 1$. Hence, we have two distinct inputs, -1 an 2 with the same output, 1, so F is not one-to-one and, hence, not invertible.

   (b) To graph F, we plot the points in F as shown in Figure 5.6.14. We see the horizontal line $y = 1$ crosses the graph more than once. Hence, the graph of F fails the Horizontal Line Test.

4. Like the function F above, the function G is described as a set of ordered pairs. Before we set about determining whether or not G is one-to-one, we take a moment to show G is, in fact, a function. That is, we must show that each real number input to G is matched

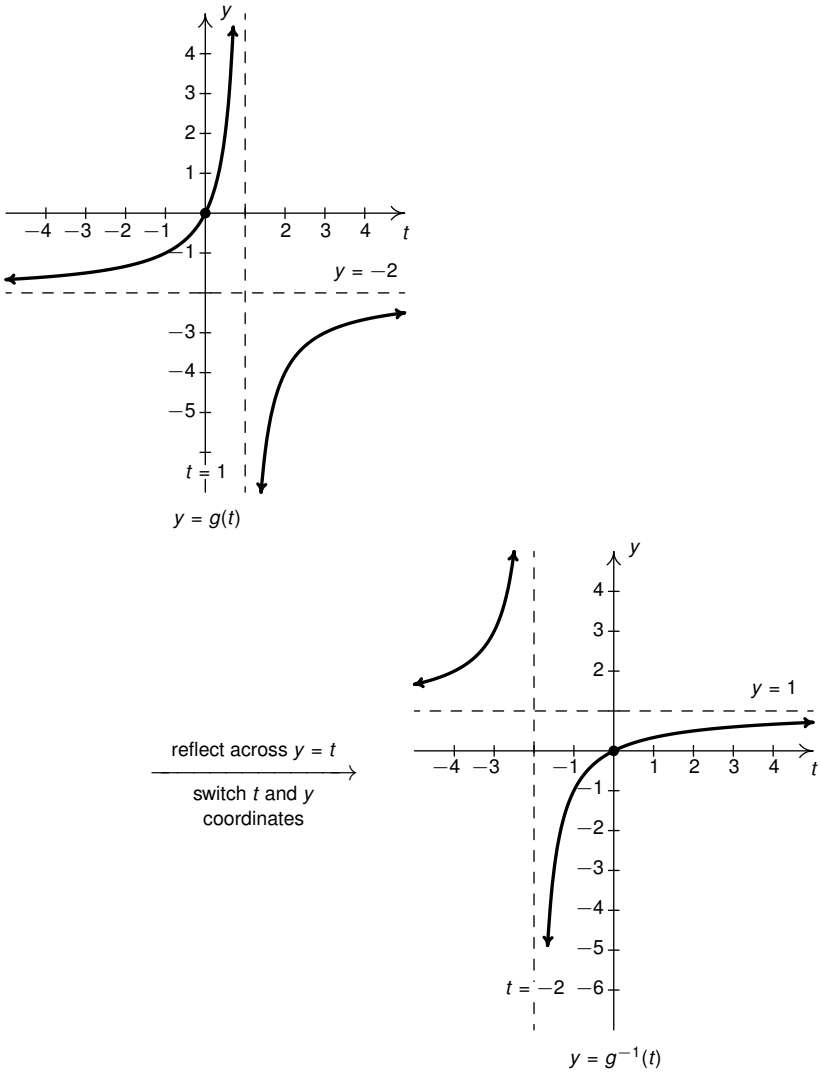

y
4
3
2
1
−4 −3 −2 −1
2 3 4
t
−1
y = −2
−3
−4
−5
t = 1
y = g(t)
reflect across y = t
switch t and y
coordinates
y
4
3
2
y = 1
−4 −3
−1
1 2 3 4
t
−1
−2
−3
−4
−5
t = −2
−6
y = g^{−1}(t)

Figure 5.6.13

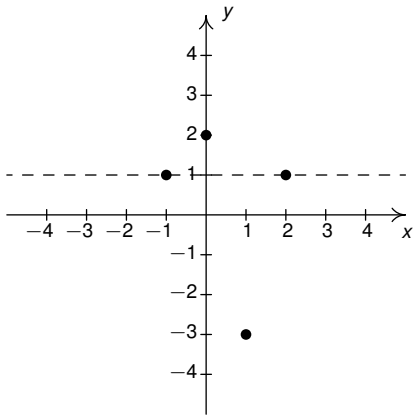


Figure 5.6.14: $y = F(x)$

to only one output.

We are given $G = \{(t^3 + 1, 2t) \mid t \text{ is a real number.}\}$. and we know that when represented in this way, each ordered pair is of the form (input, output). Hence, the inputs to G are of the form $t^3 + 1$ and the outputs from G are of the form $2t$. To establish G is a function, we must show that each input produces only one output. If it should happen that $a^3 + 1 = b^3 + 1$, then we must show $2a = 2b$. The equation $a^3 + 1 = b^3 + 1$ gives $a^3 = b^3$, or $a = b$. From this it follows that $2a = 2b$ so G is a function.

(a) To show G is one-to-one, we must show that if two outputs from G are the same, the corresponding inputs must also be the same. That is, we must show that if $2a = 2b$, then $a^3 + 1 = b^3 + 1$. We see almost immediately that if $2a = 2b$ then $a = b$ so $a^3 + 1 = b^3 + 1$ as required. This shows G is one-to-one and, hence, invertible.

(b) We graph G in Figure 5.6.15 by plotting points in the default xy-plane by choosing different values for t. For instance, $t = 0$ corresponds to the point $(0^3 + 1, 2(0)) = (1, 0)$, $t = 1$ corresponds to the point $(1^3+1, 2(1)) = (2, 2)$, $t = -1$ corresponds to

the point $((-1)^3+1, 2(-1)) = (0, -2)$, etc.[6] Our graph appears to pass the Horizontal Line Test, confirming G is one-to-one. We obtain the graph of G^{-1} below on the right by reflecting the graph of G about the line $y = x$.

□

In Example 5.6.2, we showed the functions G and g are invertible and graphed their inverses. While graphs are perfectly fine representations of functions, we have seen where they aren't the most accurate. Ideally, we would like to represent G^{-1} and g^{-1} in the same manner in which G and g are presented to us. The key to doing this is to recall that inverse functions take outputs back to their associated inputs.

Consider $G = \{(t^3+1, 2t) \mid t \text{ is a real number.}\}$. As mentioned in Example 5.6.2, the ordered pairs which comprise G are in the form (input, output). Hence to find a compatible description for G^{-1}, we simply interchange the expressions in each of the coordinates to obtain $G^{-1} = \{(2t, t^3 + 1) \mid t \text{ is a real number.}\}$.

Since the function g was defined in terms of a formula we would like to find a formula representation for g^{-1}. We apply the same logic as above. Here, the input, represented by the independent variable t, and the output, represented by the dependent variable y, are related by the equation $y = g(t)$. Hence, to exchange inputs and outputs, we interchange the 't' and 'y' variables. Doing so, we obtain the equation $t = g(y)$ which is an *implicit* description for g^{-1}. Solving for y gives an explicit formula for g^{-1}, namely $y = g^{-1}(t)$. We demonstrate this technique below.

$$\begin{aligned} y &= g(t) \\ y &= \frac{2t}{1-t} \\ t &= \frac{2y}{1-y} && \text{(interchange variables: } t \text{ and } y\text{)} \end{aligned}$$

[6] Foreshadowing Section **??**, we could let $x = t^3 + 1$ so that $t = \sqrt[3]{x-1}$. Hence, $y = 2t = 2\sqrt[3]{x-1}$.

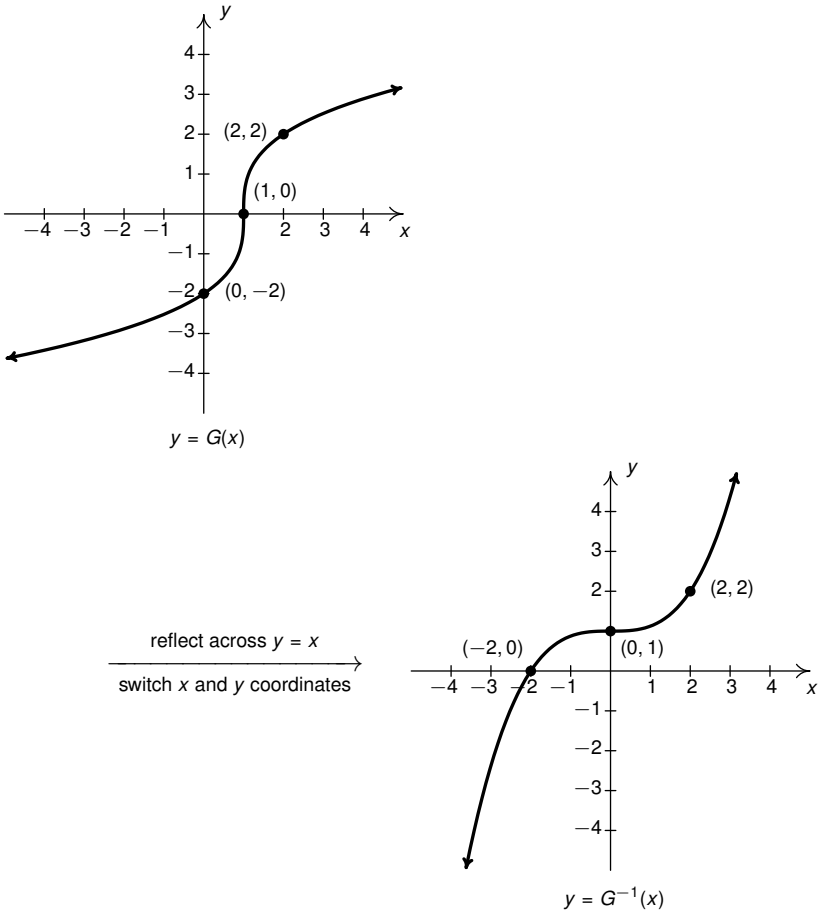

y
4
3
2
1
(2, 2)
(1, 0)
−4 −3 −2 −1
2 3 4
x
−1
−2
(0, −2)
−3
−4
y = G(x)
reflect across y = x
switch x and y coordinates
y
4
3
2
(2, 2)
(−2, 0)
(0, 1)
−4 −3 −2 −1
1 2 3 4
x
−1
−2
−3
−4
y = G^−1(x)

Figure 5.6.15

$$\begin{aligned} t(1-y) &= 2y \\ t - ty &= 2y \\ t &= ty + 2y \\ t &= y(t+2) && \text{(factor)} \\ y &= \frac{t}{t+2} \end{aligned}$$

We claim $g^{-1}(t) = \frac{t}{t+2}$, and leave the algebraic verification of this to the reader.

We generalize this approach in Box 5.6.1. As always, we resort to the default 'x' and 'y' labels for the independent and dependent variables, respectively.

Box 5.6.1: Steps for finding a formula for the Inverse of a one-to-one function

1. Write $y = f(x)$
2. Interchange x and y
3. Solve $x = f(y)$ for y to obtain $y = f^{-1}(x)$

We now return to $f(x) = x^2$. We know that f is not one-to-one, and thus, is not invertible, but our goal here is to see what way to see what goes wrong algebraically.

If we attempt to follow the algorithm above to find a formula for $f^{-1}(x)$, we start with the equation $y = x^2$ and interchange the variables 'x' and 'y' to produce the equation $x = y^2$. Solving for y gives $y = \pm\sqrt{x}$. It's this '$\pm$' which is causing the problem for us since this produces *two* y-values for any $x > 0$. See Figure 5.6.16.

Using the language of Section **??**, the equation $x = y^2$ implicitly defines *two* functions, $g_1(x) = \sqrt{x}$ and $g_2(x) = -\sqrt{x}$, each of which represents the top and bottom halves, respectively, of the graph of $x = y^2$. See Figure 5.6.17 and Figure 5.6.18.

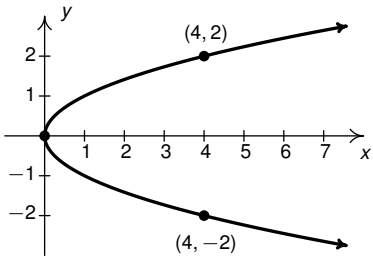


Figure 5.6.16: $x = y^2$

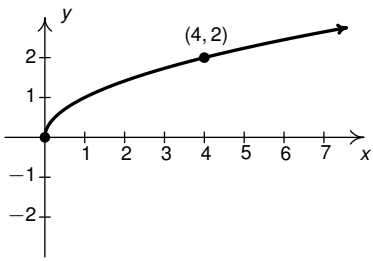


Figure 5.6.17: $y = g_1(x) = \sqrt{x}$

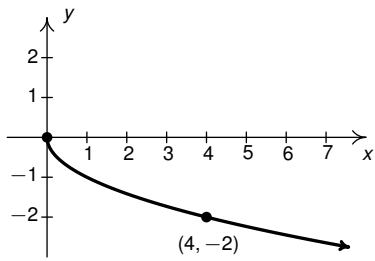


Figure 5.6.18: $y = g_2(x) = -\sqrt{x}$

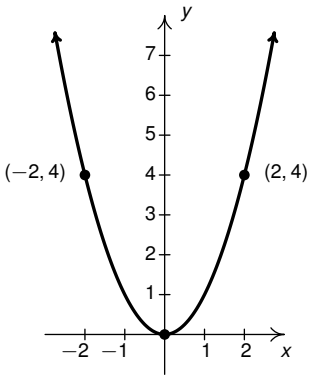


Figure 5.6.19: $y = f(x) = x^2$

Hence, in some sense, we have two *partial* inverses for $f(x) = x^2$ (shown in Figure 5.6.19): $g_1(x) = \sqrt{x}$ (shown in Figure 5.6.20) returns the *positive* inputs from f and $g_2(x) = -\sqrt{x}$ (shown in Figure 5.6.21) returns the *negative* inputs to f. In order to view each of these functions as strict inverses, however, we need to split f into two parts: $f_1(x) = x^2$ for $x \geq 0$ and $f_2(x) = x^2$ for $x \leq 0$.

We claim that f_1 and g_1 are an inverse function pair as are f_2 and g_2. Indeed, we find in Figure 5.6.22 and Figure 5.6.23.

Hence, by restricting the domain of f we are able to produce invertible functions. Said differently, in much the same way the equation $x = y^2$ implicitly describes a pair of *functions*, the equation $y = x^2$ implicitly describes a pair of *invertible* functions.

Our next example continues the theme of restricting the domain of a function to find inverse functions.

Example 5.6.3. Graph the following functions to show they are one-to-one and find their inverses. Check your answers analytically using function composition and graphically.

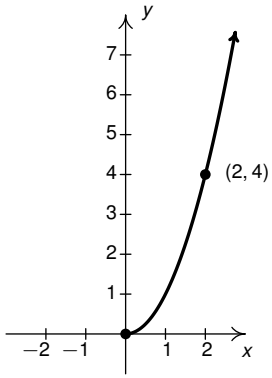


Figure 5.6.20: $y = f_1(x) = x^2$, $x \geq 0$

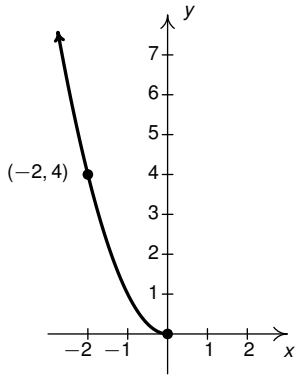


Figure 5.6.21: $y = f_2(x) = x^2$, $x \leq 0$

$$
\begin{aligned}
(g_1 \circ f_1)(x) &= g_1(f_1(x)) \\
&= g_1(x^2) \\
&= \sqrt{x^2} \\
&= |x| = x, \quad (\text{as } x \geq 0)
\end{aligned}
$$

$$
\begin{aligned}
(f_1 \circ g_1)(x) &= f_1(g_1(x)) \\
&= f_1(\sqrt{x}) \\
&= (\sqrt{x})^2 \\
&= x
\end{aligned}
$$

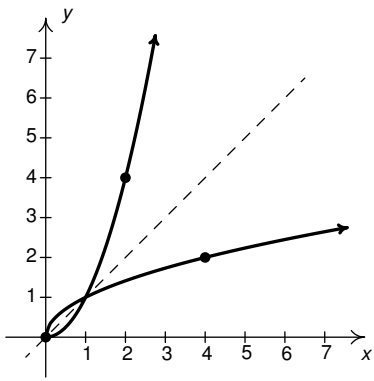
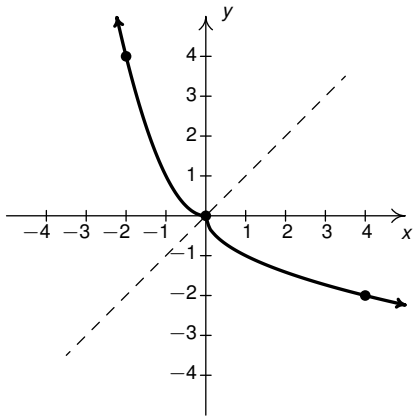


Figure 5.6.22: $y = f_1(x) = x^2$, $x \geq 0$ and $y = g_1(x) = \sqrt{x}$

$$
\begin{aligned}
(g_2 \circ f_2)(x) &= g_2(f_2(x)) \\
&= g_2(x^2) \\
&= -\sqrt{x^2} \\
&= -|x| \\
&= -(-x) = x \\
&\qquad\qquad \text{(as } x \leq 0\text{)}
\end{aligned}
$$



$$
\begin{aligned}
(f_2 \circ g_2)(x) &= f_2(g_2(x)) \\
&= f_2(-\sqrt{x}) \\
&= (-\sqrt{x})^2 \\
&= (\sqrt{x})^2 \\
&= x
\end{aligned}
$$

Figure 5.6.23: $y = f_2(x) = x^2$, $x \leq 0$ and $y = g_2(x) = -\sqrt{x}$

1. $j(x) = x^2 - 2x + 4$, $x \leq 1$.
2. $k(t) = \sqrt{t+2} - 1$

Solution.

1. The function j is a restriction of the function f from Example 5.6.2. Since the domain of j is restricted to $x \leq 1$, we are selecting only the 'left half' of the parabola. Hence, the graph of j, seen in Figure 5.6.24, passes the Horizontal Line Test and thus j is invertible. Below, we find an explicit formula for $j^{-1}(x)$ using our standard algorithm.[7]

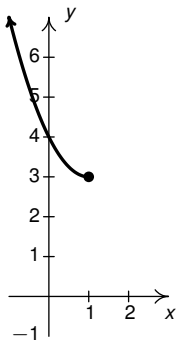


Figure 5.6.24: $y = j(x)$

$$
\begin{aligned}
y &= j(x) \\
y &= x^2 - 2x + 4, \quad x \leq 1 \\
x &= y^2 - 2y + 4, \quad y \leq 1 && \text{(switch } x \text{ and } y\text{)} \\
0 &= y^2 - 2y + 4 - x \\
y &= \frac{2 \pm \sqrt{(-2)^2 - 4(1)(4-x)}}{2(1)} && \text{(quadratic formula, } c = 4 - x\text{)}
\end{aligned}
$$

[7] Here, we use the Quadratic Formula to solve for y. For 'completeness,' we note you can (and should!) also consider solving for y by 'completing' the square.

$$
\begin{aligned}
y &= \frac{2 \pm \sqrt{4x - 12}}{2} \\
y &= \frac{2 \pm \sqrt{4(x-3)}}{2} \\
y &= \frac{2 \pm 2\sqrt{x-3}}{2} \\
y &= \frac{2\left(1 \pm \sqrt{x-3}\right)}{2} \\
y &= 1 \pm \sqrt{x-3} \\
y &= 1 - \sqrt{x-3} && \text{(since } y \leq 1\text{)}
\end{aligned}
$$

Hence, $j^{-1}(x) = 1 - \sqrt{x-3}$.

To check our answer algebraically, we simplify $(j^{-1} \circ j)(x)$ and $(j \circ j^{-1})(x)$ Note the importance of the domain restriction $x \leq 1$ when simplifying $(j^{-1} \circ j)(x)$.

$$
\begin{aligned}
\left(j^{-1} \circ j\right)(x) &= j^{-1}(j(x)) \\
&= j^{-1}\left(x^2 - 2x + 4\right), \quad x \leq 1 \\
&= 1 - \sqrt{\left(x^2 - 2x + 4\right) - 3} \\
&= 1 - \sqrt{x^2 - 2x + 1} \\
&= 1 - \sqrt{(x-1)^2} \\
&= 1 - |x - 1| \\
&= 1 - (-(x-1)) && \text{(since } x \leq 1\text{)} \\
&= x \checkmark
\end{aligned}
$$

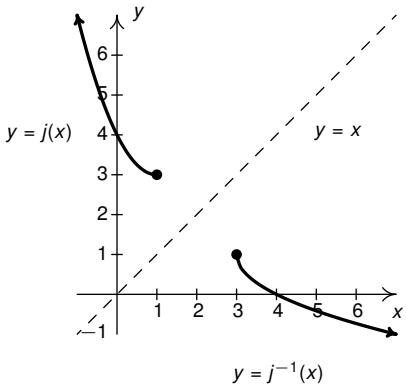


Figure 5.6.25

$$
\begin{aligned}
\left(j \circ j^{-1}\right)(x) &= j\left(j^{-1}(x)\right) \\
&= j\left(1 - \sqrt{x-3}\right) \\
&= \left(1 - \sqrt{x-3}\right)^2 - 2\left(1 - \sqrt{x-3}\right) + 4 \\
&= 1 - 2\sqrt{x-3} + \left(\sqrt{x-3}\right)^2 - 2 + 2\sqrt{x-3} + 4 \\
&= 1 + x - 3 - 2 + 4 \\
&= x \checkmark
\end{aligned}
$$

We graph both j and j^{-1} on the axes in Figure 5.6.25. They appear to be symmetric about the line $y = x$.

2. Graphing $y = k(t) = \sqrt{t+2} - 1$ (Figure 5.6.26), we see k is one-to-one, so we proceed to find an formula for k^{-1}.

$$y = k(t)$$

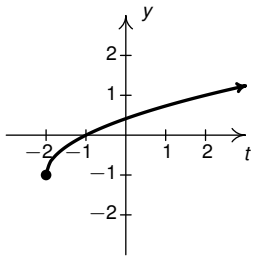


Figure 5.6.26: $y = k(t)$

$$
\begin{aligned}
y &= \sqrt{t+2} - 1 \\
t &= \sqrt{y+2} - 1 && \text{(switch } t \text{ and } y\text{)} \\
t+1 &= \sqrt{y+2} \\
(t+1)^2 &= \left(\sqrt{y+2}\right)^2 \\
t^2 + 2t + 1 &= y + 2 \\
y &= t^2 + 2t - 1
\end{aligned}
$$

We have $k^{-1}(t) = t^2 + 2t - 1$. Based on our experience, we know something isn't quite right. We determined k^{-1} is a quadratic function, and we have seen several times in this section that these are not one-to-one unless their domains are suitably restricted.

Theorem 5.6.1 tells us that the domain of k^{-1} is the range of k. From the graph of k, we see that the range is $[-1, \infty)$, which means we restrict the domain of k^{-1} to $t \geq -1$.

We now check that this works in our compositions. Note the importance of the domain restriction, $t \geq -1$ when simplifying $(k \circ k^{-1})(t)$.

$$
\begin{aligned}
\left(k^{-1} \circ k\right)(t) &= k^{-1}(k(t)) \\
&= k^{-1}\left(\sqrt{t+2} - 1\right)
\end{aligned}
$$

$$
\begin{aligned}
&= \left(\sqrt{t+2}-1\right)^2 + 2\left(\sqrt{t+2}-1\right) - 1 \\
&= \left(\sqrt{t+2}\right)^2 - 2\sqrt{t+2} + 1 + 2\sqrt{t+2} - 2 - 1 \\
&= t + 2 - 2 \\
&= t \checkmark
\end{aligned}
$$

$$
\begin{aligned}
\left(k \circ k^{-1}\right)(t) &= k\left(t^2 + 2t - 1\right), \quad t \geq -1 \\
&= \sqrt{\left(t^2 + 2t - 1\right) + 2} - 1 \\
&= \sqrt{t^2 + 2t + 1} - 1 \\
&= \sqrt{(t+1)^2} - 1 \\
&= |t+1| - 1 \\
&= t + 1 - 1, \text{ since } t \geq -1 \\
&= t \checkmark
\end{aligned}
$$

Graphically, everything checks out, provided that we remember the domain restriction on k^{-1} means we take the right half of the parabola. See Figure 5.6.27.

□

Our last example of the section gives an application of inverse functions. Recall in Example **??** in Section **??**, we modeled the demand for PortaBoy game systems as the price per system, $p(x)$ as a function of the number of systems sold, x. In the following example, we find $p^{-1}(x)$ and interpret what it means.

Example 5.6.4. Recall the price-demand function for PortaBoy game systems is modeled by the formula $p(x) = -1.5x + 250$ for $0 \leq x \leq 166$ where x represents the number of systems sold (the demand) and $p(x)$ is the price per system, in dollars.

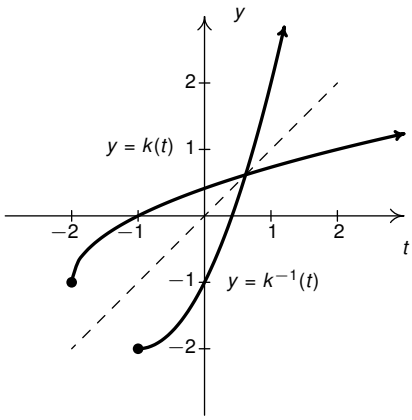


Figure 5.6.27

1. Explain why p is one-to-one and find a formula for $p^{-1}(x)$. State the restricted domain.
2. Find and interpret $p^{-1}(220)$.
3. Recall from Section **??** that the profit P, in dollars, as a result of selling x systems is given by $P(x) = -1.5x^2 + 170x - 150$. Find and interpret $\left(P \circ p^{-1}\right)(x)$.
4. Use your answer to part 3 to determine the price per PortaBoy which would yield the maximum profit. Compare with Example **??**.

Solution.

1. Recall the graph of $p(x) = -1.5x + 250$, $0 \leq x \leq 166$, is a line segment from $(0, 250)$ to $(166, 1)$, and as such passes the Horizontal Line Test. Hence, p is one-to-one. We find the expression for $p^{-1}(x)$ as usual and get $p^{-1}(x) = \frac{500-2x}{3}$. The domain of p^{-1} should match the range of p, which is $[1, 250]$, and as such, we restrict the domain of p^{-1} to $1 \leq x \leq 250$.
2. We find $p^{-1}(220) = \frac{500-2(220)}{3} = 20$. Since the function p took as inputs the number of systems sold and returned the price per system as the output, p^{-1} takes the price per system as its input and

returns the number of systems sold as its output. Hence, $p^{-1}(220) = 20$ means 20 systems will be sold in if the price is set at \$220 per system.

3. We compute $\left(P \circ p^{-1}\right)(x) = P\left(p^{-1}(x)\right) = P\left(\frac{500-2x}{3}\right) = -1.5\left(\frac{500-2x}{3}\right)^2 + 170\left(\frac{500-2x}{3}\right) - 150$. After a hefty amount of Elementary Algebra,[8] we obtain $\left(P \circ p^{-1}\right)(x) = -\frac{2}{3}x^2 + 220x - \frac{40450}{3}$.

   To understand what this means, recall that the original profit function P gave us the profit as a function of the number of systems sold. The function p^{-1} gives us the number of systems sold as a function of the price. Hence, when we compute $(P \circ p^{-1})(x) = P(p^{-1}(x))$, we input a price per system, x into the function p^{-1}.

   The number $p^{-1}(x)$ is the number of systems sold at that price. This number is then fed into P to return the profit obtained by selling $p^{-1}(x)$ systems. Hence, $\left(P \circ p^{-1}\right)(x)$ gives us the profit (in dollars) as a function of the price per system, x.

4. We know from Section **??** that the graph of $y = \left(P \circ p^{-1}\right)(x)$ is a parabola opening downwards. The maximum profit is realized at the vertex. Since we are concerned only with the price per system, we need only find the x-coordinate of the vertex. Identifying $a = -\frac{2}{3}$ and $b = 220$, we get, by the Vertex Formula, Equation **??**, $x = -\frac{b}{2a} = 165$.

   Hence, weekly profit is maximized if we set the price at \$165 per system. Comparing this with our answer from Example **??**, there is a slight discrepancy to the tune of \$0.50. We leave it to the reader to balance the books appropriately. □

[8] It is good review to actually do this!

5.6.1 Exercises

In Exercises 1 - 8, verify the given pairs of functions are inverses algebraically and graphically.

1. $f(x) = 2x + 7$ and $g(x) = \dfrac{x-7}{2}$
2. $f(x) = \dfrac{5-3x}{4}$ and $g(x) = -\dfrac{4}{3}x + \dfrac{5}{3}$.
3. $f(t) = \dfrac{5}{t-1}$ and $g(t) = \dfrac{t+5}{t}$
4. $f(t) = \dfrac{t}{t-1}$ and $g(t) = f(t) = \dfrac{t}{t-1}$
5. $f(x) = \sqrt{4-x}$ and $g(x) = -x^2 + 4$, $x \geq 0$
6. $f(x) = 1 - \sqrt{x+1}$ and $g(x) = x^2 - 2x$, $x \leq 1$.
7. $f(t) = (t-1)^3 + 5$ and $g(t) = \sqrt[3]{t-5} + 1$
8. $f(t) = -\sqrt[4]{t-2}$ and $g(t) = t^4 + 2$, $t \leq 0$.

In Exercises 9 - 28, show that the given function is one-to-one and find its inverse. Check your answers algebraically and graphically. Verify the range of the function is the domain of its inverse and vice-versa.

9. $f(x) = 6x - 2$
10. $f(x) = 42 - x$
11. $g(t) = \dfrac{t-2}{3} + 4$
12. $g(t) = 1 - \dfrac{4+3t}{5}$
13. $f(x) = \sqrt{3x-1} + 5$
14. $f(x) = 2 - \sqrt{x-5}$
15. $g(t) = 3\sqrt{t-1} - 4$
16. $g(t) = 1 - 2\sqrt{2t+5}$
17. $f(x) = \sqrt[5]{3x-1}$
18. $f(x) = 3 - \sqrt[3]{x-2}$

19. $g(t) = t^2 - 10t,\ t \geq 5$

20. $g(t) = 3(t+4)^2 - 5,\ t \leq -4$

21. $f(x) = x^2 - 6x + 5,\ x \leq 3$

22. $f(x) = 4x^2 + 4x + 1,\ x < -1$

23. $g(t) = \dfrac{3}{4-t}$

24. $g(t) = \dfrac{t}{1-3t}$

25. $f(x) = \dfrac{2x-1}{3x+4}$

26. $f(x) = \dfrac{4x+2}{3x-6}$

27. $g(t) = \dfrac{-3t-2}{t+3}$

28. $g(t) = \dfrac{t-2}{2t-1}$

29. Explain why each set of ordered pairs below represents a one-to-one function and find the inverse.

    (a) $F = \{(0,0), (1,1), (2,-1), (3,2), (4,-2), (5,3), (6,-3)\}$

    (b) $G = \{(0,0), (1,1), (2,-1), (3,2), (4,-2), (5,3), (6,-3), ...\}$

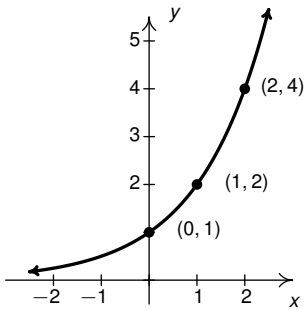
    NOTE: The difference between F and G is the '....'

    (c) $P = \{(2t^5, 3t-1) \mid t \text{ is a real number.}\}$

    (d) $Q = \{(n, n^2) \mid n \text{ is a } \textit{natural} \text{ number.}\}$[9]

In Exercises 30 - 33, explain why each graph represents[10] a one-to-one function and graph its inverse.

30. $y = f(x)$

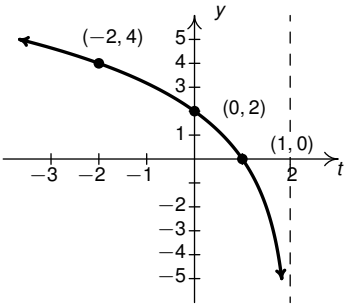


Asymptote: $y = 0$.

[9] Recall this means $n = 0, 1, 2, ...$

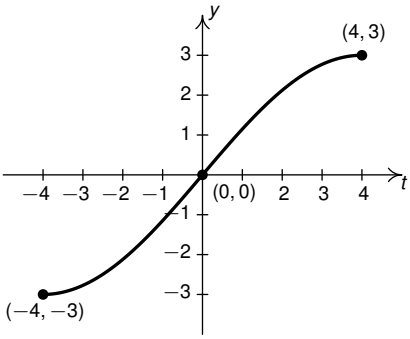
[10] or, more precisely, *appears* to represent . . .

31. $y = g(t)$



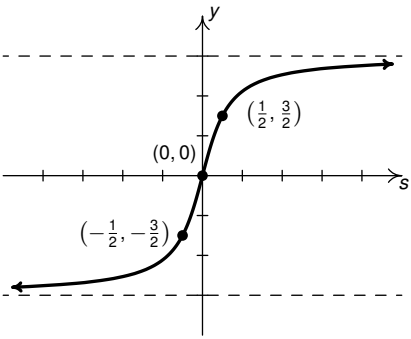
Asymptote: $t = 2$.

32. $y = S(t)$



Domain: $[-4, 4]$.

33. $y = R(s)$



Asymptotes: $y = \pm 3$.

34. The price of a dOpi media player, in dollars per dOpi, is given as a function of the weekly sales x according to the formula $p(x) = 450 - 15x$ for $0 \leq x \leq 30$.

    (a) Find $p^{-1}(x)$ and state its domain.

    (b) Find and interpret $p^{-1}(105)$.

    (c) The profit (in dollars) made from producing and selling x dOpis per week is given by the formula $P(x) = -15x^2 + 350x - 2000$, for $0 \leq x \leq 30$. Find $\left(P \circ p^{-1}\right)(x)$ and determine what price per dOpi would yield the maximum profit. What is the maximum profit? How many dOpis need to be produced and sold to achieve the maximum profit?

35. Show that the Fahrenheit to Celsius conversion function found in Exercise **??** in Section **??** is invertible and that its inverse is the Celsius to Fahrenheit conversion function.

36. Analytically show that the function $f(x) = x^3 + 3x + 1$ is one-to-one. Use Theorem 5.6.1 to help you compute $f^{-1}(1)$, $f^{-1}(5)$, and $f^{-1}(-3)$. What happens when you attempt to find a formula for $f^{-1}(x)$?

37. Let $f(x) = \dfrac{2x}{x^2 - 1}$.

    (a) Graph $y = f(x)$ using the techniques in Section **??**. Check your answer using a graphing utility.

    (b) Verify that f is one-to-one on the interval $(-1, 1)$.

    (c) Use the procedure outlined on Page 43 to find the formula for $f^{-1}(x)$ for $-1 < x < 1$.

    (d) Since $f(0) = 0$, it should be the case that $f^{-1}(0) = 0$. What goes wrong when you attempt to substitute $x = 0$ into $f^{-1}(x)$? Discuss with your classmates how this problem arose and possible remedies.

38. With the help of your classmates, explain why a function which is either strictly increasing or strictly decreasing on its entire domain would have to be one-to-one, hence invertible.

39. If f is odd and invertible, prove that f^{-1} is also odd.

40. Let f and g be invertible functions. With the help of your classmates show that $(f \circ g)$ is one-to-one, hence invertible, and that $(f \circ g)^{-1}(x) = (g^{-1} \circ f^{-1})(x)$.

With help from your classmates, find the inverses of the functions in Exercises 41 - 44.

41. $f(x) = ax + b,\ a \neq 0$

42. $f(x) = a\sqrt{x - h} + k,\ a \neq 0, x \geq h$

43. $f(x) = ax^2 + bx + c$ where $a \neq 0,\ x \geq -\dfrac{b}{2a}$.

44. $f(x) = \dfrac{ax + b}{cx + d}$, (See Exercise 45 below.)

45. What conditions must you place on the values of a, b, c and d in Exercise 44 in order to guarantee that the function is invertible?

46. The function given in number 4 is an example of a function which is its own inverse.

    (a) Algebraically verify every function of the form: $f(x) = \dfrac{ax + b}{cx - a}$ is its own inverse.

    What assumptions do you need to make about the values of a, b, and c?

    (b) Under what conditions is $f(x) = mx + b$, $m \neq 0$ its own inverse? Prove your answer.

5.6.2 Answers

9. $f^{-1}(x) = \dfrac{x+2}{6}$

10. $f^{-1}(x) = 42 - x$

11. $g^{-1}(t) = 3t - 10$

12. $g^{-1}(t) = -\frac{5}{3}t + \frac{1}{3}$

13. $f^{-1}(x) = \frac{1}{3}(x-5)^2 + \frac{1}{3}$, $x \geq 5$

14. $f^{-1}(x) = (x-2)^2 + 5$, $x \leq 2$

15. $g^{-1}(t) = \frac{1}{9}(t+4)^2 + 1$, $t \geq -4$

16. $g^{-1}(t) = \frac{1}{8}(t-1)^2 - \frac{5}{2}$, $t \leq 1$

17. $f^{-1}(x) = \frac{1}{3}x^5 + \frac{1}{3}$

18. $f^{-1}(x) = -(x-3)^3 + 2$

19. $g^{-1}(t) = 5 + \sqrt{t+25}$

20. $g^{-1}(t) = -\sqrt{\frac{t+5}{3}} - 4$

21. $f^{-1}(x) = 3 - \sqrt{x+4}$

22. $f^{-1}(x) = -\frac{\sqrt{x}+1}{2}$, $x > 1$

23. $g^{-1}(t) = \dfrac{4t-3}{t}$

24. $g^{-1}(t) = \dfrac{t}{3t+1}$

25. $f^{-1}(x) = \dfrac{4x+1}{2-3x}$

26. $f^{-1}(x) = \dfrac{6x+2}{3x-4}$

27. $g^{-1}(t) = \dfrac{-3t-2}{t+3}$

28. $g^{-1}(t) = \dfrac{t-2}{2t-1}$

29. (a) None of the first coordinates of the ordered pairs in F are repeated, so F is a function and none of the second coordinates of the ordered pairs of F are repeated, so F is one-to-one. $F^{-1} = \{(0,0), (1,1), (-1,2), (2,3), (-2,4), (3,5), (-3,6)\}$

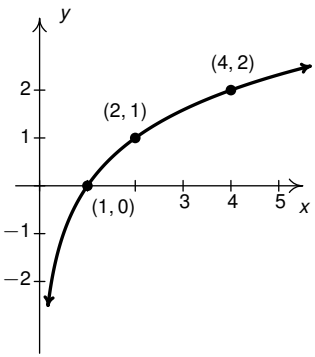
    (b) Because of the '...' it is helpful to determine a formula for the matching. For the even numbers n, $n = 0, 2, 4, ...$, the ordered pair $\left(n, -\frac{n}{2}\right)$ is in G. For the odd numbers $n = 1, 3, 5, ...$, the ordered pair $\left(n, \frac{n+1}{2}\right)$ is in G. Hence, given any input to G, n, whether it be even or odd, there is only one output from G, either $-\frac{n}{2}$ or $\frac{n+1}{2}$, both of which are functions of n. To show G is one to one, we note that if the output from G is 0 or less, then

it must be of the form $-\frac{n}{2}$ for an even number n. Moreover, if $-\frac{n}{2} = -\frac{m}{2}$, then $n = m$. In the case we are looking at outputs from G which are greater than 0, then it must be of the form $\frac{n+1}{2}$ for an odd number n. In this, too, if $\frac{n+1}{2} = \frac{m+1}{2}$, then $n = m$. Hence, in any case, if the outputs from G are the same, then the inputs to G had to be the same so G is one-to-one and $G^{-1} = \{(0,0), (1,1), (-1,2), (2,3), (-2,4), (3,5), (-3,6), ...\}$

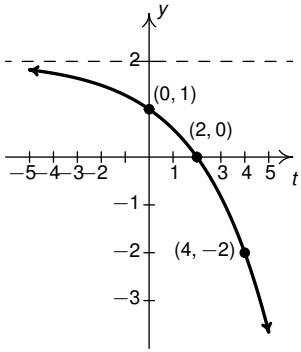
(c) To show P is a function we note that if we have the same inputs to P, say $2t^5 = 2u^5$, then $t = u$. Hence the corresponding outputs, $2t - 1$ and $3u - 1$, are equal, too. To show P is one-to-one, we note that if we have the same outputs from P, $3t - 1 = 3u - 1$, then $t = u$. Hence, the corresponding inputs $2t^5$ and $2u^5$ are equal, too. Hence P is one-to-one and $P^{-1} = \{(3t - 1, 2t^5) \mid t \text{ is a real number.}\}$

(d) To show Q is a function, we note that if we have the same inputs to Q, say $n = m$, then the outputs from Q, namely n^2 and m^2 are equal. To show Q is one-to-one, we note that if we get the same output from Q, namely $n^2 = m^2$, then $n = \pm m$. However since n and m are *natural* numbers, both n and m are positive so $n = m$. Hence Q is one-to-one and $Q^{-1} = \{(n^2, n) \mid n \text{ is a } \textit{natural} \text{ number.}\}$.

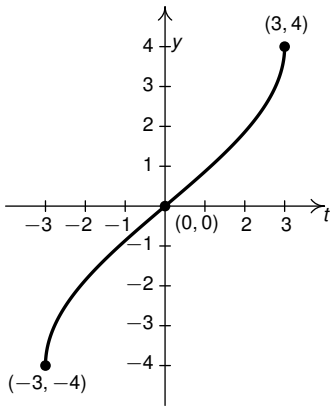
30. $y = f^{-1}(x)$. Asymptote: $x = 0$.



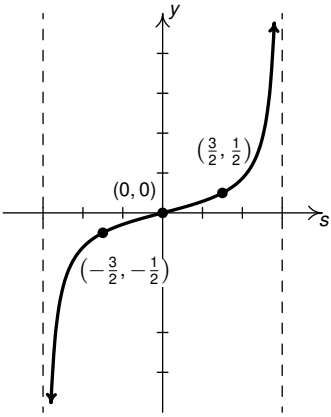
31. $y = g^{-1}(t)$. Asymptote: $y = 2$.



32. $y = S^{-1}(t)$. Domain $[-3, 3]$.



33. $y = R^{-1}(s)$. Asymptotes: $s = \pm 3$.



34. (a) $p^{-1}(x) = \frac{450-x}{15}$. The domain of p^{-1} is the range of p which is $[0, 450]$

    (b) $p^{-1}(105) = 23$. This means that if the price is set to \$105 then 23 dOpis will be sold.

    The graph of $y = \left(P \circ p^{-1}\right)(x)$ is a parabola opening downwards with vertex $\left(275, \frac{125}{3}\right) \approx (275, 41.67)$. This means that the maximum profit is a whopping \$41.67 when the price per dOpi is set to \$275. At this price, we can produce and sell $p^{-1}(275) = 11.\overline{6}$ dOpis. Since we cannot sell part of a system, we need to adjust the price to sell either 11 dOpis or 12 dOpis. We find $p(11) = 285$ and $p(12) = 270$, which means we set the price per dOpi at either \$285 or \$270, respectively. The profits at these prices are $\left(P \circ p^{-1}\right)(285) = 35$ and $\left(P \circ p^{-1}\right)(270) = 40$, so it looks as if the maximum profit is \$40 and it is made by producing and selling 12 dOpis a week at a price of \$270 per dOpi.

36. Given that $f(0) = 1$, we have $f^{-1}(1) = 0$. Similarly $f^{-1}(5) = 1$ and $f^{-1}(-3) = -1$

46. (b) If $b = 0$, then $m = \pm 1$. If $b \neq 0$, then $m = -1$ and b can be any real number.